

# Cybersecurity Incident Report

## Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: **During the initial phase of the incident, the backend infrastructure provided standard [SYN, ACK] responses to the adversary's synchronization requests, as evidenced in log entries 52 through 54. Subsequently, the external actor deviated from normative traffic patterns by transmitting a continuous and excessive volume of SYN packets to the target system.**

The logs show that: **The presence of an HTTP/1.1 504 Gateway Time-out (text/html) notification signifies that a gateway server failed to receive a timely response from the backend web server. When processing delays exceed established thresholds, the intermediary infrastructure transmits this error to the client browser.**

**Furthermore, logs indicate [RST, ACK] packets, which serve to reset and acknowledge a terminated attempt. These occur when the expected [SYN, ACK] transmission fails to reach the web server, forcing the visitor's browser to drop the connection and display a timeout error. Users may subsequently attempt a new SYN request by refreshing their interface.**

This event could be: **This occurrence is identified as a SYN flood event. Within this specific threat scenario, malicious actors exploit the TCP framework to saturate the target infrastructure by transmitting an overwhelming volume of synchronization requests.**

## Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- 1. The [SYN] packet is the initial request from an employee visitor trying to connect to a web page hosted on the web server. SYN stands for "synchronize."**
- 2. The [SYN, ACK] packet is the web server's response to the visitor's request agreeing to the connection. The server will reserve system resources for the final step of the handshake. SYN, ACK stands for "synchronize acknowledge."**
- 3. The [ACK] packet is the visitor's machine acknowledging the permission to connect. This is the final step required to make a successful TCP connection. ACK**

stands for “acknowledge.”

Explain what happens when a malicious actor sends a large number of SYN packets all at once: **Threat actors may exploit the TCP framework by saturating a target with SYN requests during the initial handshake phase. Should the volume of these requests exceed the system's capacity to process them, the infrastructure becomes congested and ceases responding. This represents a network-layer denial of service (DoS) event, specifically a SYN flood, which consumes available bandwidth to impede legitimate traffic. Such attacks mimic TCP setups to overwhelm the recipient with data packets. A direct DoS involves one originating point, whereas a distributed denial of service (DDoS) utilizes various sources across disparate regions, complicating the identification of the adversary.**

Explain what the logs indicate and how that affects the server: **An HTTP/1.1 504 Gateway Time-out (text/html) error message. This message is generated by a gateway server that was waiting for a response from the web server. If the web server takes too long to respond, the gateway server will send a timeout error message to the requesting browser. An [RST, ACK] packet, which would be sent to the requesting visitor if the [SYN, ACK] packet is not received by the web server. RST stands for reset, acknowledge. The visitor will receive a timeout error message in their browser and the connection attempt is dropped. The visitor can refresh their browser to attempt to send a new SYN request.**